

Four Possible Research Directions in Quantum VRP

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1 Qubit Efficient VRP Solver

Traditional quantum VRP solvers often use a straightforward but inefficient qubit allocation method. The common approach allocates N^2 qubits, where N is the number of customers. This is done to encode both the nodes and the sequence of their visits.

Let's assume $N = 6$, which requires 36 qubits and a 236-dimensional state space. Each qubit q_{ij} represents whether the route includes a directed edge from node i to node j . For example, in a system with four nodes, qubit $q_{1,2}$ encodes whether the vehicle travels from location 1 to location 2. The final route is interpreted as a path composed of edges where the corresponding qubits are in the $|1\rangle$ state.

Research Direction

A promising research direction is to adopt a **node-visit-time encoding**, a more qubit-efficient method inspired by research like the work from [1]. This approach aims to significantly reduce the number of qubits required, making it feasible to solve larger VRP instances on current and near-future quantum computers.

2 VRP Preprocessing Challenge

VRPs typically involve spatial points (depots, customers) and road networks. A crucial preprocessing step is to construct a complete graph from these points by finding the shortest path between all pairs of locations on the road network. There are two primary classical approaches for this:

1. **All-Pairs Shortest Paths (APSP):** Calculate the shortest paths for all pairs of nodes in the road network and then extract the subset relevant to the depots and customers. Best classical algorithms for this include Floyd-Warshall or min-plus matrix multiplication.

2. **Single-Source Shortest Path (SSSP):** Calculate the shortest paths from each depot and customer to all other depots and customers. A* is a common algorithm for this.

Research Direction

A research direction here is to explore the use of **quantum distributed algorithms** [2] to compute APSP. The findings can be compared with the SSSP Bellman-Ford algorithm, which could potentially be accelerated using **quantum dynamic programming** [3].

3 Managing Constraints in VRP

Constraints (e.g., traffic, time windows, vehicle capacity) are central to VRP. A naive approach is to handle these during preprocessing by modifying graph weights. For example, a longer but less congested road might be given a lower weight than a shorter but traffic-heavy one. This results in a modified graph that can be solved with standard VRP methods.

Research Direction

An alternative, inspired by guided local search methods, is to manage constraints iteratively during the optimization process. In the context of quantum computing, a research direction is to explore how to incorporate constraints into **Quantum Approximate Optimization Algorithm (QAOA)**.

- **Penalty in Cost Hamiltonian:** The conventional approach is to add a penalty term to the cost Hamiltonian for states that violate constraints [4].
- **Constraint in Mixer Hamiltonian:** A more advanced approach is to design a mixer Hamiltonian that only allows transitions between states that satisfy the constraints.
- **Adaptive QAOA Steps:** Another method could involve adding more QAOA steps to an iteration, allowing the algorithm to satisfy constraints dynamically as it converges to an optimal solution [5].

4 Swap Test for Clustering

VRPs can involve a vast number of data points, making clustering a necessary preprocessing step to reduce the problem size. Classical clustering algorithms like k-means can be computationally intensive, with a complexity of $O(N \cdot k)$ and a faster version using k-d trees at $O(N \log N)$. HPC research has further parallelized k-d trees to achieve $O(N \log N/p)$ speedup.

Research Direction

A quantum research direction is to use **angle encoding** to represent 2D spatial points as a 2-qubit quantum state. Once the data is encoded in this manner, a **quantum swap test** can be used to calculate the inner product between different data points. This allows for a form of **quantum parallelism** [6] to cluster the data more efficiently than classical methods. This could be a powerful tool for reducing the complexity of large-scale VRPs.

References

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